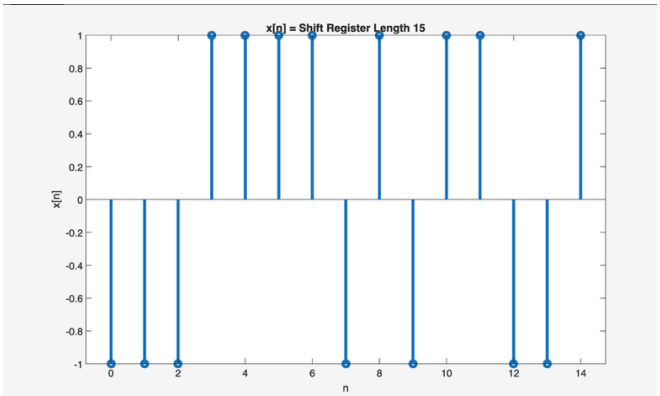
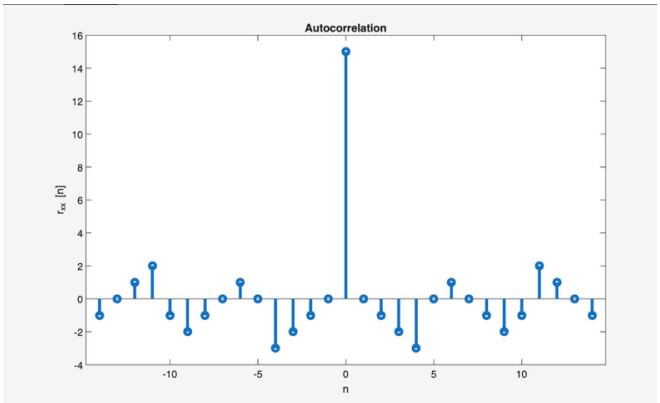
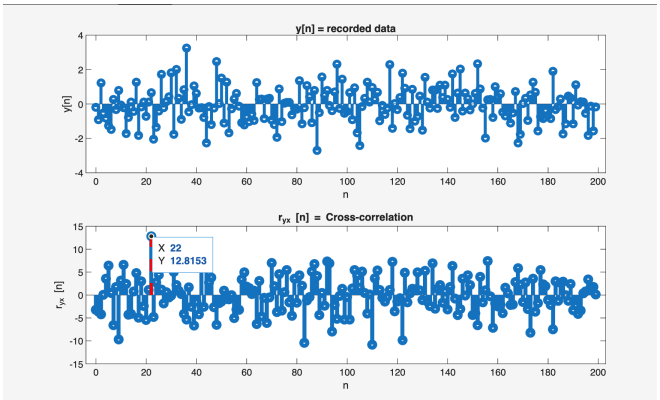
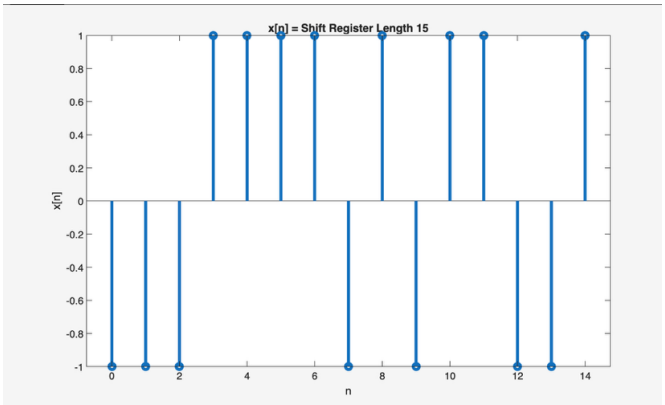
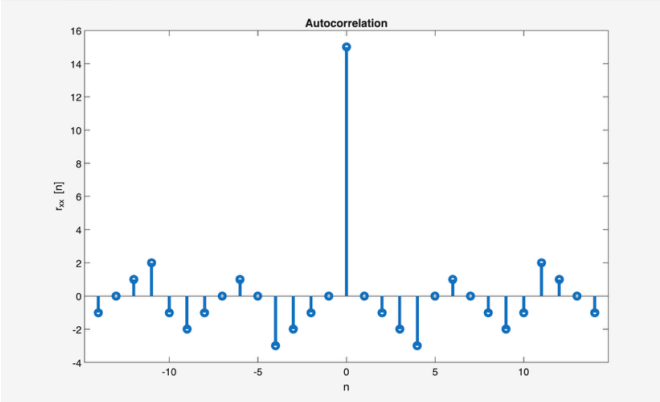
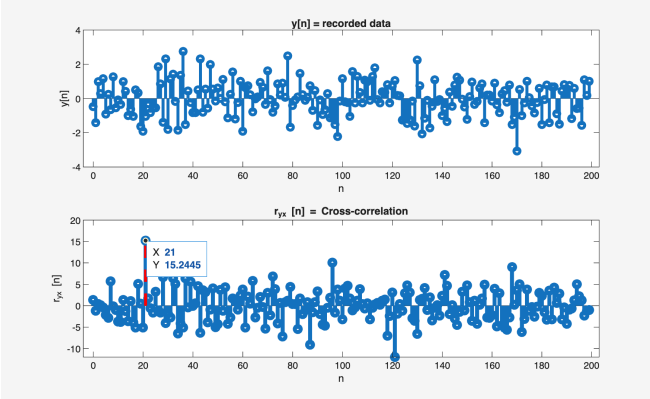
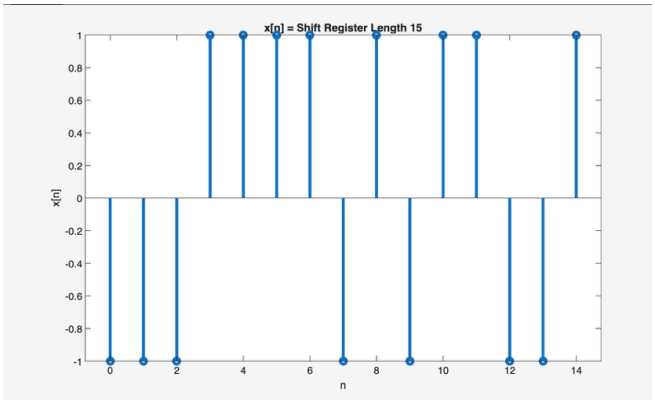
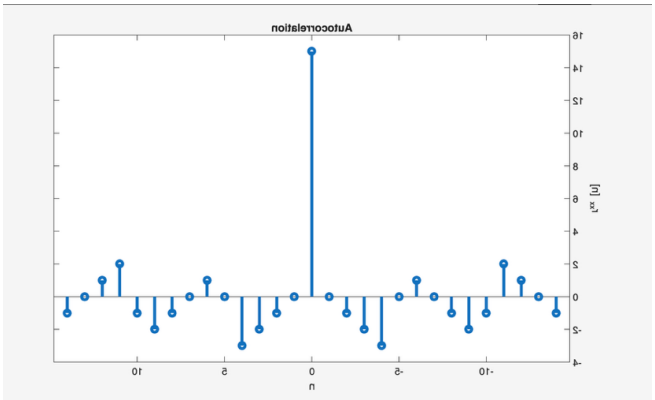
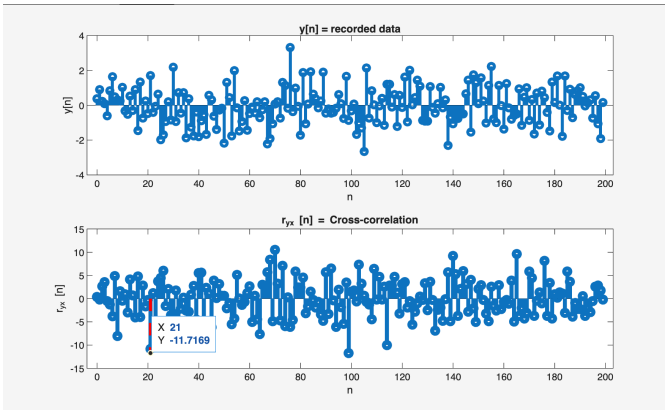
Javier Aldana

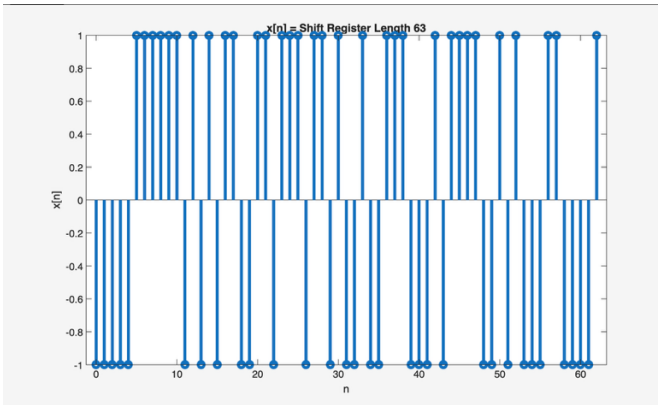
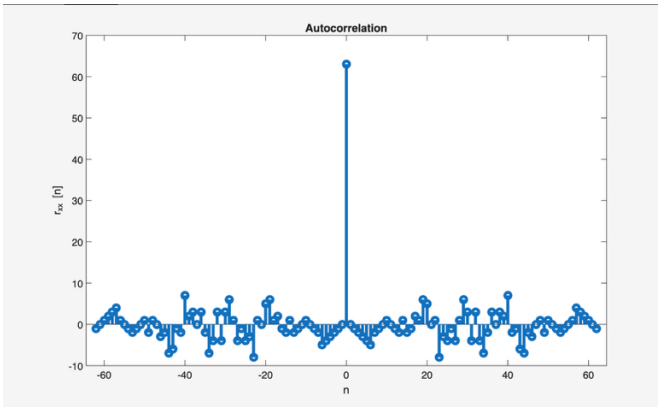
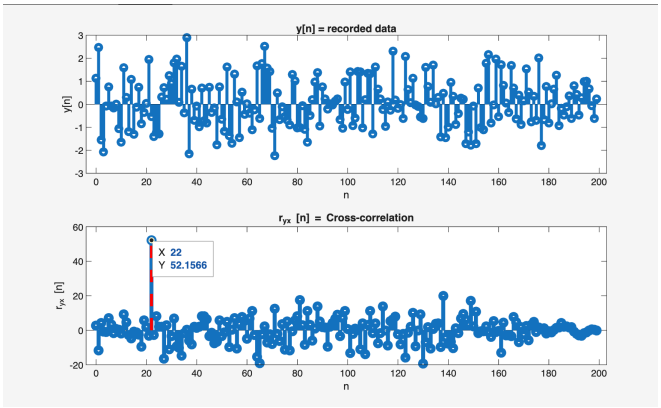
ECE 538 Assignment 1

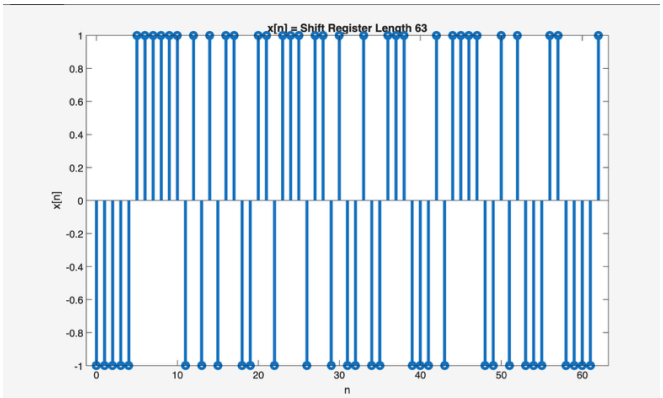
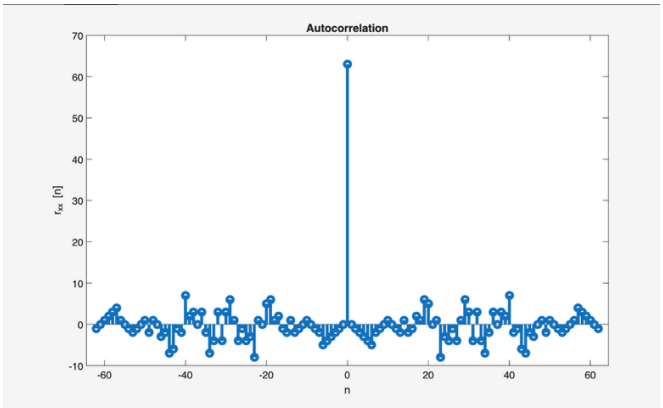
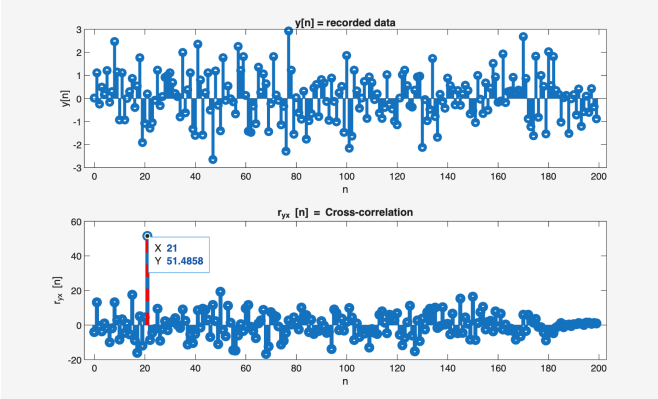
9/21/2026

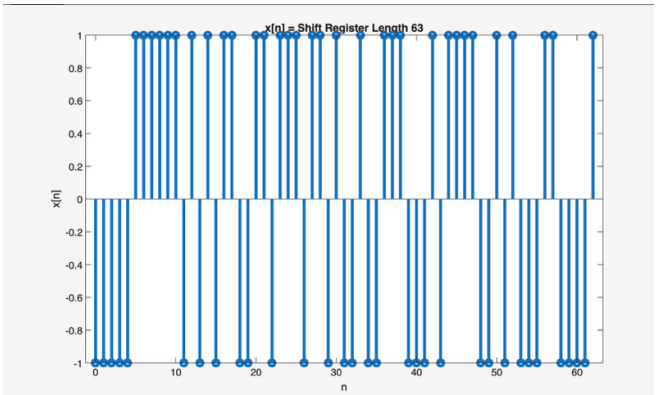
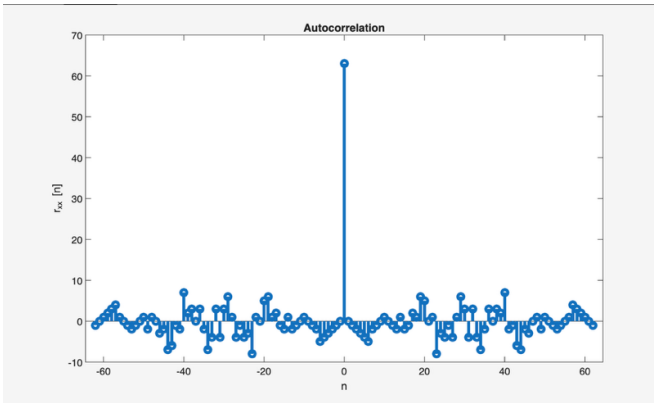
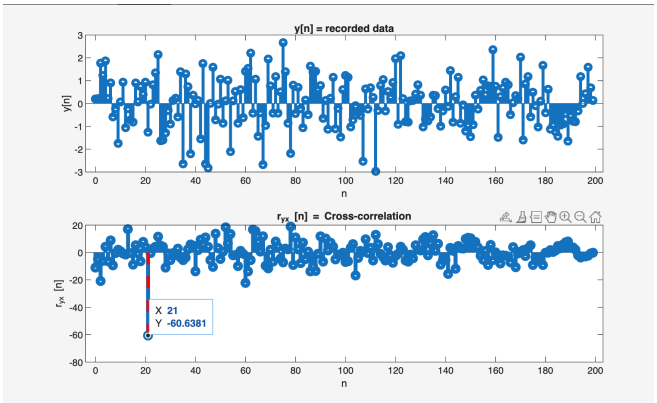
#	Chart	Description
1i.		Figure 7i. is the shift register of length 127, it was made in the M127.m file.
1ii.		Figure 7ii. is the autocorrelation of the shift register of length 127.
1iii.		Figure 7iii. is the recorded y[n] = x[n-20] + x[n-22] + v[n] and the cross correlation of y[n] with the code. There is a much higher peak at x = 22 for the cross-correlation graph. The magnitude is close to the expected value of 15 but there are a lot of other deltas that are close to a magnitude of 15.

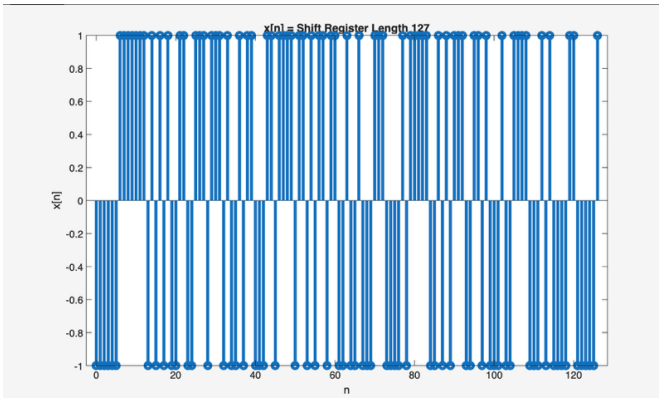
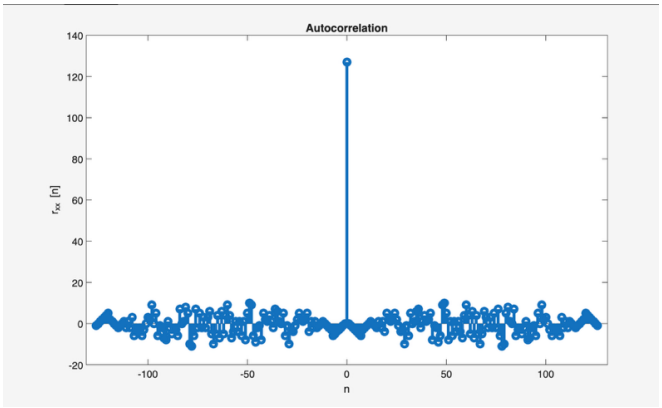
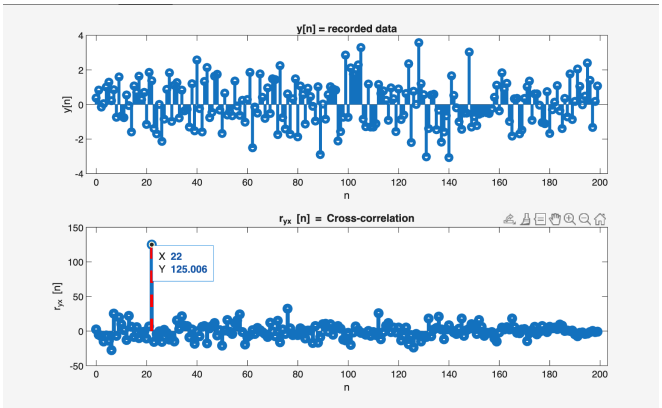
#	Chart	Description
2i.		Figure 2i. is the same shift register of length 15, it was made in the M15.m file.
2ii.		Figure 2ii. is the autocorrelation of the shift register.
2iii.		Figure 2iii. is the recorded y[n] = x[n-20] + x[n-21] + v[n] and the cross correlation of y[n] with the code. There is a visible peak at x = 21 for the cross-correlation graph. There could be other peaks at 120 and 95, it is hard to tell.

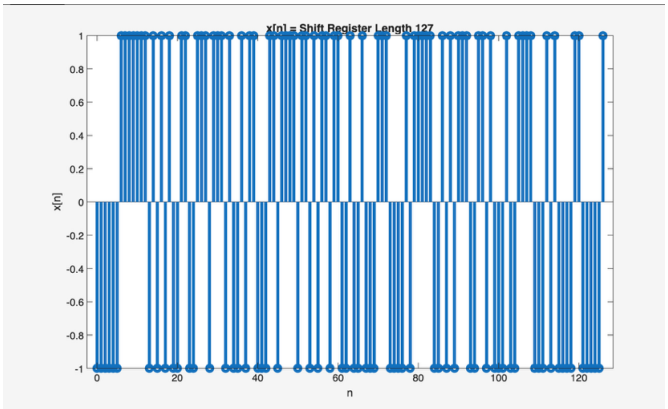
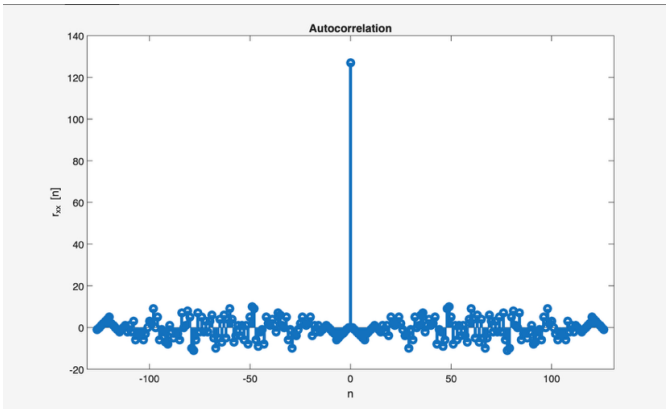
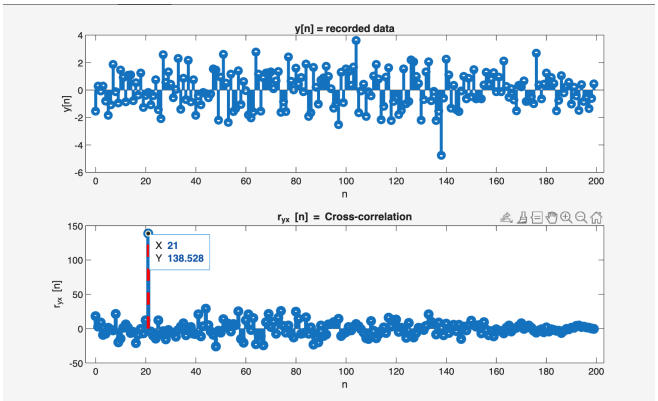
#	Chart	Description
3i.		Figure 3i. is the same shift register of length 15, it was made in the M15.m file.
3ii.		Figure 3ii. is the autocorrelation of the shift register.
3iii.		Figure 3iii. is the recorded y[n] = x[n-20] + -x[n-21] + v[n] and the cross correlation of y[n] with the code. There is a visible but small negative peak at x = 21 for the cross-correlation graph. There is another higher magnitude peak at 98 and 66.

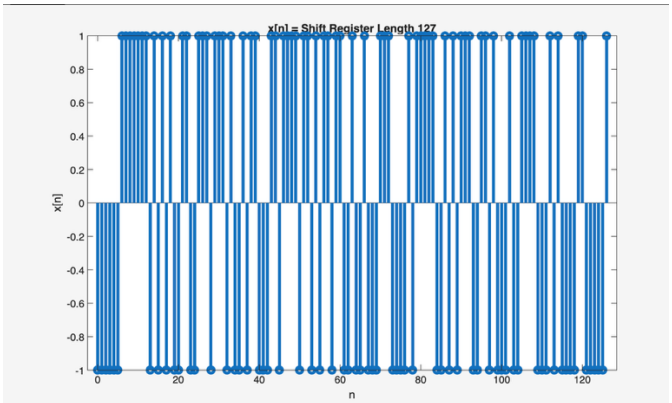
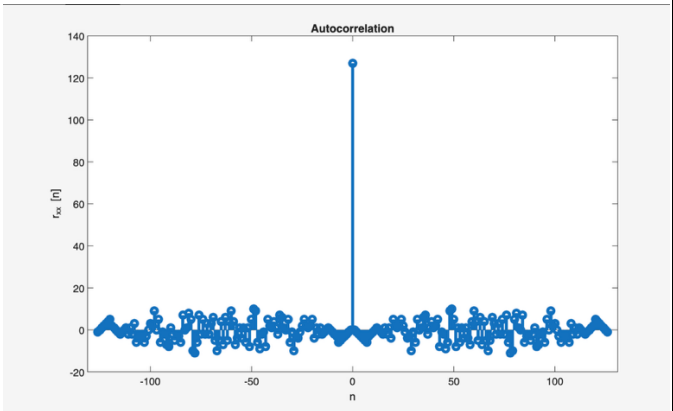
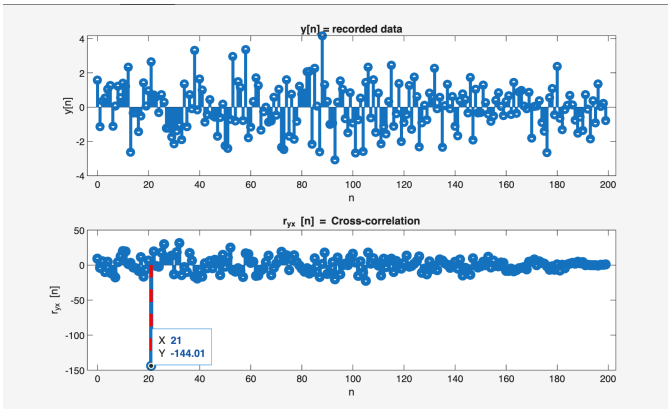
#	Chart	Description
4i.		Figure 4i. is the shift register of length 63, it was made in the M63.m file.
4ii.		Figure 4ii. is the autocorrelation of the shift register of length 63.
4iii.		Figure 4iii. is the recorded y[n] = x[n-20] + x[n-22] + v[n] and the cross correlation of y[n] with the code. There is a much higher peak at x = 22 for the cross-correlation graph. There are no other visible peaks.

#	Chart	Description
5i.		Figure 5i. is the same shift register of length 63, it was made in the M63.m file.
5ii.		Figure 5ii. is the autocorrelation of the shift register of length 63.
5iii.		Figure 4iii. is the recorded y[n] = x[n-20] + x[n-22] + v[n] and the cross correlation of y[n] with the code. There is a much visible peak at x = 21 for the cross-correlation graph.

#	Chart	Description
6i.		Figure 6i. is the shift register of length 63, it was made in the M63.m file.
6ii.		Figure 6ii. is the autocorrelation of the shift register of length 63.
6iii.		Figure 6iii. is the recorded y[n] = x[n-20] - x[n-22] + v[n] and the cross correlation of y[n] with the code. There is a very high magnitude peak of close to 63 at x = 21 for the cross-correlation graph.

#	Chart	Description
7i.		Figure 7i. is the shift register of length 127, it was made in the M127.m file.
7ii.		Figure 7ii. is the autocorrelation of the shift register of length 127.
7iii.		Figure 7iii. is the recorded y[n] = x[n-20] + x[n-22] + v[n] and the cross correlation of y[n] with the code. There is a peak of around 127 at x = 22 for the cross-correlation graph.

#	Chart	Description
8i.		Figure 8i. is the shift register of length 127, it was made in the M127.m file.
8ii.		Figure 8ii. is the autocorrelation of the shift register of length 127.
8iii.		Figure 8iii. is the recorded y[n] = x[n-20] + x[n-22] + v[n] and the cross correlation of y[n] with the code. There is a much higher peak at x = 21 for the cross-correlation graph.

#	Chart	Description
9i.		Figure 9i. is the shift register of length 127, it was made in the M127.m file.
9ii.		Figure 9ii. is the autocorrelation of the shift register of length 127.
9iii.		Figure 9iii. is the recorded y[n] = x[n-20] - x[n-21] + v[n] and the cross correlation of y[n] with the code. There is a very high negative peak at x = 21 for the cross-correlation graph. However it is significantly lower than 127, this could be due to noise randomness.

CODE:

SimpleCrossCorrEG.m

```
clear all
clf
%D1=7;
D2=21;
code = M63();
disp(code);
%Remember:  n=0 corresponds to 1st location in array
%0 cannot be an index of an array
M=length(code);
NL=200;  %"arbitrary" length
codeplusw=randn(1,NL);
%codeplusw(D1+1:D1+M)=codeplusw(D1+1:D1+M)+code;
codeplusw(D2+1:D2+M)= codeplusw(D2+1:D2+M) - code;
ryx=conv(codeplusw,fliplr(code));
ydata=ryx(M:M+NL-1);
%throw away first M-1 values, IMPORTANT ^^
figure(1)
stem([0:M-1],code,'LineWidth',3)
rxx=conv(code,fliplr(code));
title('x[n] = Shift Register Length 15')
xlabel('n')
ylabel('x[n]')
figure(2)
stem([-(M-1):(M-1)],rxx,'LineWidth',3)
title('Autocorrelation')
xlabel('n')
ylabel('r_{xx} [n]')
figure(3)
subplot(2,1,1)
stem(0:NL-1,codeplusw,'LineWidth',3)
title('y[n] = recorded data')
xlabel('n')
ylabel('y[n]')
subplot(2,1,2)
stem(0:NL-1,ydata,'LineWidth',4)
title('r_{yx} [n] = Cross-correlation')
xlabel('n')
ylabel('r_{yx} [n]')
hold on
%plot([D1 D1], [0 max((ydata))],'r--','LineWidth',3)
plot([D2 D2], [0 -max(abs(ydata))],'r--','LineWidth',3)
hold off
```

M15.m

```
function [m15] = M15()

clear all
clf
register=[1 0 0 0];
for ri=1:15,
m15(ri)=register(1,4);
register(2:4)=register(1:3);
register(1,1)=rem((register(1,1)+m15(1,ri)),2);
end
m15=2*m15-1;
stem((0:14),m15)
```

M63.m

```
function [m63] = M63()

clear all
clf
register=[1 0 0 0 0 0];
for ri=1:63,
m63(ri)=register(1,6);
register(2:6)=register(1:5);
register(1,1)=rem((register(1,1)+m63(1,ri)),2);
end
m63=2*m63-1;
stem((0:62),m63)
```

M127.m

```
function [m127] = M127()

clear all
clf

    register=[1 0 0 0 0 0 0];
    for ri=1:127,
    m127(ri)=register(1,7);
    register(2:7)=register(1:6);
    register(1,1)=rem((register(1,1)+m127(1,ri)),2);
    end
    m127=2*m127-1;
    stem((0:126),m127)

end
```